AI and Data Literacy

ECON [TBD] — Fall 2027

Lectures: Tuesdays & Thursdays, [TBD]
Section: once a week, [TBD]
Classroom: [TBD]

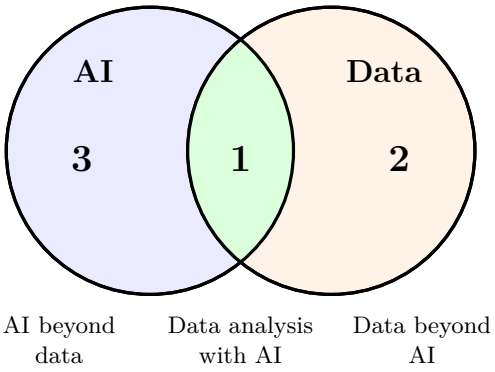
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Teaching Assistant: [TBD]

1 Course Description

AI has driven the cost of producing a data analysis to nearly zero. A question that used to take a trained analyst weeks to months of work, collecting the data, cleaning it, writing the code, running the statistical tests, interpreting the results, and writing up the report, now takes an agent minutes. What AI did not make cheap is knowing whether the analysis is right and understanding what it means. That is what this course teaches.

The course is built for every Baylor first-year, whatever you plan to major in. The next person to analyze data will not be a statistician. It will be a nurse, a coach, a pastor, a reporter, a teacher, a manager. This course is for the student you already are, not an attempt to turn you into a quant.

The course has three parts, which you can picture as two overlapping circles, AI and Data:



1. **Data analysis with AI.** Where the circles overlap. You will use an AI agent (a program that reads your files, writes and runs code, and reports back to you in plain English) to analyze real data, following a checklist, in weekly sections and in three empirical projects.
2. **Data beyond AI.** The statistics you need to check an agent's work: distributions, sampling and uncertainty, regression, prediction, and what fabricated data looks like.
3. **AI beyond data.** What AI is and how language models work; what machines did to work in the first and second machine ages; who builds AI and who governs it; and what it means for your career.

Lectures cover parts 2 and 3. Sections and projects cover part 1. The whole course runs on one loop, practiced every week: the agent *produces*; you *verify*; you *understand*.

2 Course Objectives

By the end of the semester you will be able to:

1. **Direct** an AI agent to turn a question into an analysis of real data, following a defined checklist from raw data to figures, tables, and estimates.
2. **Check** an analysis against what the numbers should look like: read a histogram, a summary table, or a regression table and say whether it is plausible, where it might be wrong, and what would be a sign of error or fabrication.
3. **Explain** what an analysis shows and what it does not, in plain English, to someone who did not do it: whether it describes, predicts, or claims a cause, and how uncertain it is.
4. **Know the machine**: what a language model is doing when it answers, how agents differ from chat, what this technology did to work in the past and may do to yours, and when not to trust it.

3 Course Format

Two 75-minute lectures a week and one required weekly section led by the teaching assistant.

Lectures are where you learn the statistics and the ideas about AI. They are mostly "data beyond AI" and "AI beyond data." You will take notes by hand.

Sections are where you work with the agent. You will bring a laptop, work with real data, follow the course checklist, and make progress on the current empirical project under the TA's guidance. Sections are also where project presentations happen.

4 Required Tools and Readings

- **A laptop**, required for section. In the first week you will install two free programs on it: Claude Code, the AI agent we use all semester, and VS Code, the editor that shows you the files the agent is working on.
- **A Claude subscription** (about $20/month) for the semester. If this is a financial hardship, contact me.
- **Readings**, posted on Canvas. There is *no required textbook*. Each week has one short reading, listed in the schedule. Finish it before Tuesday's class, because that is when the quiz on it happens. A few weeks use chapters of *OpenIntro Statistics*, a free online book (https://www.openintro.org/book/os/).

4.1 Books I will be drawing on

None of these are required. I will be using them myself to build the course, and you may find them useful if you want to go deeper.

- Mitchell, Melanie (2019). *Artificial Intelligence: A Guide for Thinking Humans*. Picador.
- Brynjolfsson, Erik and Andrew McAfee (2014). *The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies*. Norton.
- Goldin, Claudia and Lawrence F. Katz (2008). *The Race Between Education and Technology*. Harvard University Press.
- Garicano, Luis, Jin Li, and Yanhui Wu. *Messy Jobs: The Work That AI Cannot Reach*.
- Newport, Cal (2016). *Deep Work: Rules for Focused Success in a Distracted World*. Grand Central.

One more shaped how the course is designed rather than what it covers: Willingham, Daniel T. (2009), *Why Students Don't Like School* (Jossey-Bass). Its one-sentence lesson, that students remember what they think about, is why every assignment in this course asks you a question the agent cannot answer for you.

5 Assignments and Grading

5.1 Grade Distribution

Component	Weight	Description
Class Attendance	5%	Lectures and section; see policy below
Quizzes	20%	Eleven, in class, on the readings; two lowest dropped
Exams (3)	45%	15% each; in class; no final exam
Empirical Projects (3)	30%	10% each; Measurement, Prediction, Causality

5.2 Class Attendance (5%)

Attendance is taken at every lecture and section. The mechanism will be announced the first week. You may miss three meetings (lectures or sections) without penalty. After that, the attendance grade is capped:

Absences	Maximum attendance score
0–3	100
4	80
5	50
6 or more	0

5.3 Quizzes (20%)

A short quiz at the start of Tuesday's class in the weeks marked in the schedule, on the reading listed for that week. Closed book, no devices. There are eleven quizzes; your two lowest scores are dropped. Quizzes cannot be made up. Their job is to make sure you did the reading and to tell you, early and often, whether you are following.

5.4 Exams (45%)

Three in-class exams, each worth 15%. Each is a mix of multiple choice, fill in the blank, and short essay. Closed book, no electronics. There is no final exam. Exam 1 covers Weeks 1–5, Exam 2 covers Weeks 7–10, and Exam 3 covers Weeks 12–15. Exams test the lectures and the readings: the statistics, the stories, and the ideas about AI. If you engage seriously in class and with the readings you will do well. If you let the agent do your thinking outside of class, you will not. Each exam also asks a few questions about your most recent project: know your project well enough to answer questions about any number in it.

5.5 Empirical Projects (30%)

Three projects, each worth 10%, each done with an AI agent, each following the same rules:

- **Production, verification, and understanding.** The agent produces. You verify the output against what you know the numbers should look like, and you record what you checked. You explain what it means in your own words. Each project is graded on all three, and the last two carry most of the weight.
- **The checklist.** Each project comes with a defined list of steps I want to see done, in order. You follow it and show that you did. The first step, before the agent runs anything, is to write down what you expect the answer to look like, so that checking the output is a comparison and not a rationalization.
- **A report and a deck.** A short written report and a short slide deck, both built with the agent, both explained by you.
- **Replicability.** You download the data yourself, organize it in a project folder on your own machine with raw data kept read-only, and submit a folder from which someone else could rerun everything: data, code, output, a README (a short file saying what is in the folder), and your session logs (the saved transcript of everything you and the agent said to each other).
- **Presentation.** In the section after each project is due, students are called on at random to give a five-minute presentation of their project to the class. If you are called on and are not present or not prepared, you lose credit on the project.

Project 1: Measurement. Document patterns in the structure of wages across the United States from 2000 to the present using the American Community Survey, by race, gender, education, and college major (field of degree is available in the ACS from 2009). You will download the data from IPUMS, organize it locally, work through the checklist, interpret the results, and write them up.

Project 2: Prediction. You will choose from a small set of options, such as predicting elections or predicting recidivism (whether a person released from jail is arrested again). You will use regression, R^2, in-sample versus out-of-sample fit, covariates, overfitting, LASSO, and cross-validation, all with the agent, and explain what the model can and cannot predict.

Project 3: Causality. In 2014 a study in *Science* by LaCour and Green claimed that a short conversation could change minds about gay marriage. The data turned out to be fabricated, and two graduate students caught it with the same tools you learn in this course. You will replicate how they caught it, compare the fabricated data to the real survey it was copied from, analyze the real experiment that followed, and be able to discuss what the episode teaches about data, experiments, and trust.

5.6 Grade Scale

Percentage	Grade
90–100	A
80–89.99	B
70–79.99	C
60–69.99	D
Below 60	F

6 Course Policies

6.1 Electronics Policy

Lectures: Laptops, tablets, and phones are put away. Take notes by hand. Exceptions only for documented accommodations.

Sections: Laptops are required. Phones remain put away.

6.2 AI Policy

This course treats AI as both the subject and the tool. The rule is simple: use AI to *learn more*, not to *think less*.

- **Encouraged:** using the agent for the projects and in section, using AI to help you understand the readings, and using it to teach yourself anything the course touches.
- **Prohibited:** any AI or device during quizzes and exams, and submitting agent-written text as your own verification notes or interpretation. Concretely: code, figures, tables, and formatting may be produced by the agent. The parts of each project headed *What I checked* and *What it means*, and everything you say in your presentation, must be written by you.

The quizzes, exams, and random presentations are in the room without the agent. That is where the course finds out whether the learning happened.

6.3 Academic Integrity

Students are expected to maintain the highest standards of academic integrity. Plagiarism, cheating, and submitting AI-generated work as your own thinking will not be tolerated. Violations result in disciplinary action up to and including a failing grade. Refer to the Baylor University Academic Integrity Policy.

6.4 Accommodations

Students who require accommodations should contact the Office of Access and Learning Accommodation (OALA) at the start of the semester. Please also reach out to me so we can discuss how to support your learning.

6.5 Late Work

Late projects lose 10% per day unless prior arrangements have been made. Extensions are granted only for exceptional, documented circumstances and must be requested in advance. Quizzes and exams cannot be made up.

6.6 Communication

Check your university email regularly. Announcements will be sent by email and posted on Canvas.

7 Lauren's Promise

Lauren McCluskey, a 21-year-old honors student-athlete at the University of Utah, was murdered on October 22, 2018, by a stalker. Lauren's Promise is my pledge to listen and believe you if someone is threatening you. Any form of sexual harassment or violence will not be excused or tolerated at Baylor University.

If you are experiencing sexual assault, relationship violence, or stalking:

- If you are in immediate danger, call 911.
- Call Baylor Police Department at (254) 710-2222.
- Report it to me, and I will connect you to resources.
- Seek confidential support: Counseling (254) 710-2467; Health Services (254) 710-1010.

Course Schedule

Classes begin Tuesday, August 24 and end Tuesday, December 7. There is no class the week of Thanksgiving (November 22–26). Each week's reading is listed in its Section row; finish it before Tuesday, when the quiz on it is given. Readings are tentative and will be finalized on Canvas before each week.

Week	Date	Type	Topics
Part I: Measurement			
1	Aug 24	Lecture	**We will all be quants.** AI drove the cost of analysis to zero; what it did not make cheap is checking and understanding. Produce, verify, understand. How the course works.
	Aug 26	Lecture	**What is AI?** A short history. Machine learning, neural networks, language models, agents. What a language model is doing when it answers.
		Section	Install Claude Code and VS Code. Open a project folder. First prompts. *Reading:* Mitchell (2021), "Why AI Is Harder Than We Think."
2	Aug 31	Quiz 1 & Lecture	**Describing data.** A simple question about wages. The histogram. Mean, median, and why they differ. Spread: percentiles and the standard deviation, in pictures.
	Sep 2	Lecture	**The first machine age.** The Industrial Revolution. The weavers got rich, then the machines came for weaving. The Luddites. Wages fell for twenty years before jobs did. Ricardo changed his mind.
		Section	*Project 1 assigned.* The ACS from IPUMS: download, organize the project folder, raw data read-only. Checklist steps 1–3. *Reading:* Acemoglu and Johnson (2024), "Learning from Ricardo and Thompson."
3	Sep 7	Quiz 2 & Lecture	**Comparing groups.** Wages by race, gender, and education. Weighted means and survey weights. Why the comparison you make determines what you find.
	Sep 9	Lecture	**Sampling and uncertainty.** The miracle of polling. A sample is a set of coin flips. The sampling distribution narrows with n. Standard error, margin of error, and what a 95% interval does and does not mean.
		Section	Project 1: weighted means by group with the agent. First figures. *Reading:* OpenIntro Statistics, chapters on summarizing data and sampling (free online).
4	Sep 14	Quiz 3 & Lecture	**Do two things move together?** Covariance and correlation as pictures. Anscombe's quartet. Zero correlation is not no relationship. Correlation is not causation, with the confound named.
	Sep 16	Lecture	**Computers and work.** Routine and non-routine tasks. Computers hollowed out the middle of the labor market. Two college graduates with different fates.
		Section	Project 1: trends 2000 to present, by group. Figures that carry the argument. *Reading:* Autor (2024), "Applying AI to Rebuild Middle Class Jobs."
5	Sep 21	Quiz 4 & Lecture	**The rhetoric of data.** A figure is an argument. Tables, figures, and words that tell the truth well. What a good report and a good deck look like.

Week	Date	Type	Topics
	Sep 23	Lecture	**The second machine age.** Tasks as building blocks. Complements and substitutes. What generative AI did to productivity in the first studies. *Project 1 due.*
		Section	**Project 1 presentations**, called at random. *Reading:* Brynjolfsson, Li, and Raymond (2025), "Generative AI at Work."
6	Sep 28	Lecture	**Exam 1 review.**
	Sep 30	Exam 1	Covers Weeks 1–5.
		Section	No new material; exam debrief and Project 2 preview.
Part II: Prediction			
7	Oct 5	Quiz 5 & Lecture	**The best line through the data.** Least squares was invented to find a lost asteroid. The regression line. The coefficient on a binary variable is a difference in means. Residuals.
	Oct 7	Lecture	**Applied regression.** Women earn 84 cents on the dollar. The gap, its standard error, then add education and experience and watch it shrink. "Holding constant" changes the question. R^2.
		Section	*Project 2 assigned.* Choose your option. Download and organize the data. Checklist steps 1–3. *Reading:* OpenIntro Statistics, chapters on linear and multiple regression.
8	Oct 12	Quiz 6 & Lecture	**Same regression, different purpose.** Description versus prediction. In-sample versus out-of-sample. Add variables and R^2 climbs; add enough and it hits 1.00. Overfitting. Hold out a test set; the test set is sacred.
	Oct 14	Lecture	**How a language model works.** Tokens. Fixed weights. Next-token prediction and the draw. Attention. Why it can be confidently wrong.
		Section	Project 2: baseline regression; train and test split with the agent. *Reading:* [TBD: an accessible explainer on language models].
9	Oct 19	Quiz 7 & Lecture	**Let the data pick.** LASSO in pictures: shrink the coefficients and some go to zero. Cross-validation as taking turns. The sweet spot between underfitting and overfitting.
	Oct 21	Lecture	**Prediction in a courtroom.** COMPAS, a risk score U.S. courts use to predict rearrest. ProPublica found its errors were racially asymmetric. Northpointe's defense and the impossibility behind it. When a prediction should not be used.
		Section	Project 2: LASSO and cross-validation with the agent. Compare models out of sample. *Reading:* Angwin et al. (2016), "Machine Bias," *ProPublica.*
10	Oct 26	Quiz 8 & Lecture	**The agent can be wrong.** I asked agents to estimate the effect of the minimum wage. Some switched methods when they disliked the answer. They read their own names. The errors are not typos; they are hidden choices.
	Oct 28	Lecture	**Verification.** Design before results. Why the checklist is in that order. Verification works when errors are imperfectly correlated: a second agent, a second language, your own eyes. *Project 2 due.*

Week	Date	Type	Topics
		Section	**Project 2 presentations**, called at random. *Reading:* [TBD: Cunningham, agent hidden actions].
11	Nov 2	Lecture	**Exam 2 review.**
	Nov 4	Exam 2	Covers Weeks 7–10.
		Section	*Project 3 assigned.* The LaCour story. The three datasets: the fabricated study, the real national survey, and the real experiment. Checklist steps 1–3.
Part III: Causality			
12	Nov 9	Quiz 9 & Lecture	**What would have happened otherwise?** The causal question. Selection: people who go to hospitals are already sick. What randomization buys. The experiment as the gold standard.
	Nov 11	Lecture	**When data lies, part 1.** LaCour and Green claimed a conversation could change minds. The results were striking. Then two graduate students tried to build on it.
		Section	Project 3: compare the fabricated study's baseline to the national survey it was copied from. Histograms, means, standard deviations. *Reading:* LaCour and Green (2014, retracted); Broockman, Kalla, and Aronow (2015).
13	Nov 16	Quiz 10 & Lecture	**When data lies, part 2.** Heaping at 50, the pile-up at zero, the recipe. The real experiment, and the more modest truth. Why adding a control can change an answer: omitted variable bias. Lies leave fingerprints.
	Nov 18	Lecture	**Who builds AI.** GPUs, data centers, and compute. Training runs from 10,000 chips to 100,000. Closed models and open weights. The firms and what they want.
		Section	Project 3: the real experiment. The treatment effect as a difference in means and as a regression. *Reading:* Broockman and Kalla (2016).
14	Nov 23, 25	*No class — Thanksgiving.*	
15	Nov 30	Quiz 11 & Lecture	**Who governs AI.** Chip export controls and the geopolitics of compute. The EU AI Act and U.S. policy. Privacy. Who decides, and the Baylor line that conscience is not replaced.
	Dec 2	Lecture	**What you will be.** Careers when the next data user is everyone. Incumbents and entrants. Use agents when the work is valuable, slow, and checkable. Deskilling is real but not deterministic. Keep your hands on the wheel. *Project 3 due.*
		Section	**Project 3 presentations**, called at random. *Reading:* Acemoglu, Autor, and Johnson (2026), "Building Pro-Worker AI."
16	Dec 7	Exam 3	Covers Weeks 12–15. Last class meeting.